Phase-Locked Boundary Entanglement Engineering and Holographic Warp Metrics: A Static Holographic Boundary Theory Investigation

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I. INTRODUCTION AND HOLOGRAPHIC FRAMEWORK

Metric engineering in classical general relativity is an inverse problem. One first specifies a target Lorentzian geometry $g_{\mu\nu}^{\text{target}}$, computes its Einstein tensor, and then infers the stress-energy tensor required to sustain it:

$$T_{\mu\nu}^{\text{req}} = \frac{1}{8\pi G_N}\left(G_{\mu\nu}[g^{\text{target}}] + \Lambda g_{\mu\nu}^{\text{target}}\right). \tag{1}$$

For an Alcubierre-type longitudinal shift [2], the wall gradients of the prescribed shape function generate regions in which the Eulerian energy density is negative,

$$\rho_{\text{GR}} = T_{\mu\nu}^{\text{req}} n^\mu n^\nu < 0, \qquad \text{and hence} \qquad T_{00} < 0 \quad \text{in part of the wall.} \tag{2}$$

The obstruction is therefore not merely technological: the target metric is supported by a bulk source that violates the weak energy condition and, for appropriate null directions, the null energy condition.

Static Holographic Boundary Theory (SHBT) changes the control problem. The independent data are assigned to a static boundary conformal register, while the bulk metric is treated as a downstream projection of boundary loading, Shannon entanglement, and holographic resolution flow [3–5]. The foundational construction and its numerical implementation are established in the primary repository [1]. In the extension studied here, no negative-energy bulk source is inserted into Eq. (1). Instead, a unitary character excitation redistributes boundary entanglement and the resulting directional entropy shear is mapped into a bulk shift component. The operational distinction is

$$\text{classical GR:} \quad g_{\mu\nu}^{\text{target}} \longrightarrow T_{\mu\nu}^{\text{req}}, \qquad \text{SHBT:} \quad \rho_\partial \longrightarrow \rho_E \longrightarrow \Phi_s^{(x)} \longrightarrow g_{\mu\nu}. \tag{3}$$

Thus *exotic-matter-free metric projection* refers specifically to the absence of an imposed exotic bulk source. Control hardware and positive boundary bookkeeping costs remain part of the physical accounting.

A. Canonical anomaly-free branch

The visible boundary theory is the affine product

$$\mathcal{A}_{\text{vis}} = \mathrm{SU}(2)_{k_\ell} \times \mathrm{SU}(3)_{k_q}, \qquad \mathfrak{b}_* = (k_\ell, k_q, K) = (26, 8, 312) = (26, 8, 312), \tag{4}$$

embedded in the parent completion $\mathrm{SO}(10)_K$. The corresponding center lifts are

$$\begin{aligned} I_\ell^* &= \frac{K}{2k_\ell} = \frac{312}{52} = 6, \\ I_q^* &= \frac{K}{3k_q} = \frac{312}{24} = 13. \end{aligned} \tag{5}$$

For $\|x\|_{\mathbb{Z}} \equiv \min_{n\in\mathbb{Z}} |x - n|$, the scalar framing defect is

$$\Delta_{\text{fr}}(\mathfrak{b}) = \max\left\{\left\|\frac{K}{2k_\ell}\right\|_{\mathbb{Z}}, \left\|\frac{K}{3k_q}\right\|_{\mathbb{Z}}\right\}. \tag{6}$$

Equations (4) and (5) imply

$$\Delta_{\text{fr}}(\mathfrak{b}_*) = \max\{\|6\|_{\mathbb{Z}}, \|13\|_{\mathbb{Z}}\} = 0. \tag{7}$$

The framing condition is not an optional numerical tolerance. Within the SHBT closure ansatz, the scalar obstruction has the unique metric-proportional bulk continuation

$$\mathcal{E}_{\mu\nu} = \mathcal{Q}_{\text{fr}} \Delta_{\text{fr}} g_{\mu\nu}, \tag{8}$$

where $\mathcal{Q}_{\rm fr}$ is the framing-response coefficient. Therefore

$$\begin{aligned}\Delta_{\rm fr} = 0 \quad &\Longrightarrow \quad \mathcal{E}_{\mu\nu} = 0, \\ \|\mathcal{E}\|_g &= |\mathcal{Q}_{\rm fr}||\Delta_{\rm fr}|\|g\|_g = 0.\end{aligned} \tag{9}$$

Any intervention that changes the affine levels may reopen $\Delta_{\rm fr}$, inject a framing-sourced Weyl term, and invalidate the closed boundary-to-bulk projection. Admissible controls must consequently act inside the fixed nine-dimensional visible block and preserve the branch $\mathfrak{b}_*$ exactly.

B. Holographic control variables

The completed dark ledger and its dimensionless modular debt are

$$\begin{aligned}c_{\rm dark}^{\rm comp} &= \frac{1197103}{362670} = 3.300805139659, \\ \Delta_{\rm mod} &= \frac{c_{\rm dark}^{\rm comp}}{24} = \frac{1197103}{8704080} = 0.137533547486.\end{aligned} \tag{10}$$

The extension assigns the corresponding kinematic scale

$$v_{\rm eff} = c\exp\left(\frac{\Delta_{\rm mod}}{2}\right), \qquad \frac{v_{\rm eff}}{c} = 1.071186351. \tag{11}$$

These definitions do not by themselves establish a realizable propulsion system. They specify a closed mathematical model whose consistency can be tested by modular invariance, unitary excitation, framing closure, metric nonsingularity, and positivity of the Euclidean Gram representative. Sections II and III construct those ingredients explicitly and distinguish the positive projection metric from the Lorentzian spacetime metric into which the shift is embedded.

II. BOUNDARY REGISTER ARCHITECTURE AND CHARACTER EXCITATIONS

A. Nine-component visible modular register

The visible coordinate set is the Cartesian product of three $\mathrm{SU}(2)_{26}$ charge labels and three low $\mathrm{SU}(3)_8$ weights:

$$\mathcal{C} = \left\{c_{ij} = (i,j) \,\middle|\, q_i \in (22,23,26),\ w_j \in \big((0,0),(1,0),(0,1)\big)\right\}. \tag{12}$$

The associated visible Hilbert space and coordinate projectors are

$$\mathcal{H}_{\rm vis} = \mathrm{span}\left\{|q_i, w_j\rangle\right\}_{i,j=0}^{2}, \qquad \Pi_{ij} = |q_i, w_j\rangle\langle q_i, w_j|. \tag{13}$$

Consequently $\dim\mathcal{H}_{\rm vis} = 9$, and any visible density matrix is represented by a 9×9 positive semidefinite matrix of unit trace.

The modular loading is obtained from the visible blocks of the $\mathrm{SU}(2)_{26}$ and $\mathrm{SU}(3)_8$ modular S matrices. With the baseline charge embedding used on both SU(2) axes, the raw loading is

$$\widetilde{\rho}_{\bar{E}}(i,j) = \left|S^{\mathrm{SU}(2)_{26}}_{q_i,q_j}\right|^2 \left|S^{\mathrm{SU}(3)_8}_{w_i,w_j}\right|^2. \tag{14}$$

The SU(2) kernel is

$$S^{\mathrm{SU}(2)_{26}}_{ab} = \sqrt{\frac{2}{28}}\sin\left[\frac{\pi(a+1)(b+1)}{28}\right]. \tag{15}$$

For $\lambda = (p,q)$, define the three-vector and Weyl vector

$$v(p,q) = \frac{1}{3}\left(2p+q,\ q-p,\ -2p-q\right), \qquad \varrho = v(1,1). \tag{16}$$

The corresponding $\mathrm{SU}(3)_8$ Weyl-sum kernel is

$$S^{\mathrm{SU}(3)_8}_{\lambda\mu} = -\frac{i}{11\sqrt{3}} \sum_{\sigma\in S_3} \mathrm{sgn}(\sigma) \exp\left[-\frac{2\pi i}{11}\left(\sigma[v(\lambda)+\varrho], v(\mu)+\varrho\right)\right]. \tag{17}$$

The boundary normalization and normalized visible loading density are

$$Z_\partial = \sum_{i,j=0}^{2} \widetilde{\rho}_{\bar{E}}(i,j), \qquad \rho_{\bar{E}}(i,j) = \frac{\widetilde{\rho}_{\bar{E}}(i,j)}{Z_\partial}, \qquad \sum_{i,j=0}^{2} \rho_{\bar{E}}(i,j) = 1. \tag{18}$$

The Rust/PyO3 engine evaluates

$$Z_\partial = 0.00342010966891, \qquad \left|\sum_{i,j} \rho_{\bar{E}}(i,j) - 1\right| = 0. \tag{19}$$

The local Shannon contributions, total visible entropy, and normalized entanglement density are

$$s_{ij} = -\rho_{\bar{E}}(i,j)\ln\rho_{\bar{E}}(i,j), \qquad S_E = \sum_{i,j=0}^{2} s_{ij}, \qquad \rho_E(i,j) = \frac{s_{ij}}{S_E}. \tag{20}$$

Thus $\sum_{i,j}\rho_E(i,j) = 1$, while the computed baseline entropy is

$$S_E = 0.999490161461. \tag{21}$$

B. Phase-locked character excitation

Write each SU(3) weight as $w_j = (p_j, r_j)$ and define its scalar grade

$$\nu(w_j) = p_j + r_j. \tag{22}$$

To preserve the compact notation requested for character phases, we use $q_i + w_j$ as shorthand for $q_i + \nu(w_j)$. The explicit character modulation factor is therefore

$$\chi^{\mathrm{char}}_{ij}(\theta) = e^{i\theta(q_i+w_j)} \equiv \exp\left\{i\theta\left[q_i + \nu(w_j)\right]\right\}. \tag{23}$$

The conformal dimensions and visible central charge are

$$h_{ij} = \frac{q_i(q_i+2)}{4(26+2)} + \frac{p_j^2 + r_j^2 + p_j r_j + 3p_j + 3r_j}{3(8+3)}, \qquad c_{\mathrm{vis}} = \frac{39}{14} + \frac{64}{11} = \frac{1325}{154}. \tag{24}$$

The diagonal phase-locking operator combines the character phase with the modular T-phase:

$$M_{\mathrm{phase}}(\theta) = \sum_{i,j=0}^{2} \exp\left\{i\theta[q_i + \nu(w_j)] + 2\pi i\theta\left(h_{ij} - \frac{c_{\mathrm{vis}}}{24}\right)\right\}\Pi_{ij}. \tag{25}$$

Let $G_x = G_x^\dagger$ be the directional generator that couples adjacent charge sectors at fixed SU(3) weight. The unitary excitation operator is

$$\widehat{\mathcal{O}}_{\mathrm{excitation}}(\theta) = e^{-i\theta G_x} M_{\mathrm{phase}}(\theta), \qquad 0 \le \theta < 2\pi. \tag{26}$$

Because G_x is Hermitian and every diagonal phase has unit modulus,

$$\widehat{\mathcal{O}}^\dagger_{\mathrm{excitation}}(\theta)\widehat{\mathcal{O}}_{\mathrm{excitation}}(\theta) = M^\dagger_{\mathrm{phase}} e^{i\theta G_x} e^{-i\theta G_x} M_{\mathrm{phase}} = \mathbb{I}_9. \tag{27}$$

The numerical infinity-norm residual is

$$\left\|\widehat{\mathcal{O}}^\dagger_{\mathrm{excitation}}\widehat{\mathcal{O}}_{\mathrm{excitation}} - \mathbb{I}_9\right\|_\infty = 4.587893\times 10^{-16}. \tag{28}$$

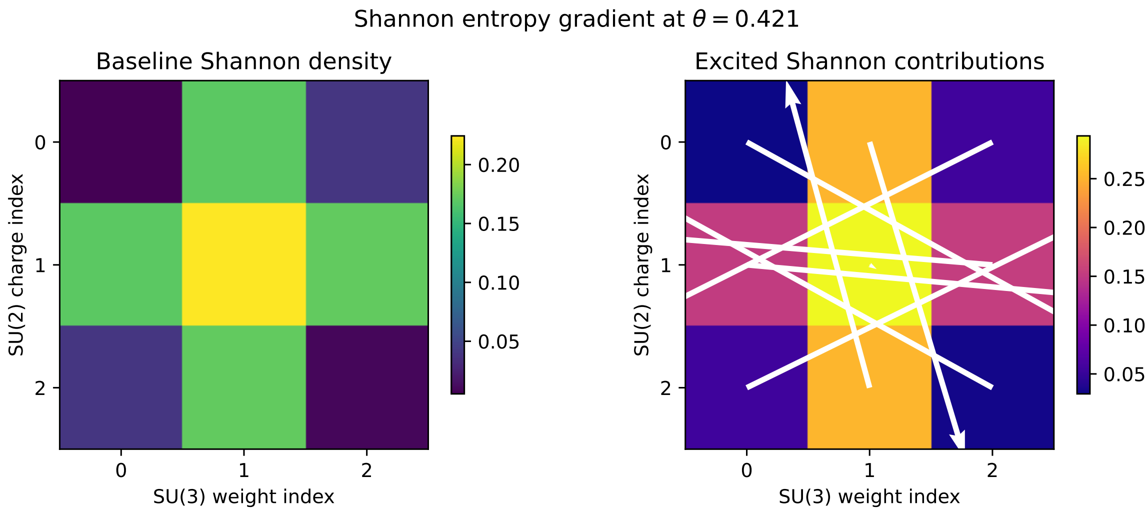

FIG. 1. Baseline and phase-locked Shannon contributions at $\theta = 0.421$. The arrows display the discrete gradient of the excited entropy contribution.

For a baseline state ρ_∂, the excited state and observable coordinate probabilities are

$$\rho_\partial^{(\theta)} = \frac{\widehat{\mathcal{O}}_{\text{excitation}}(\theta)\rho_\partial\widehat{\mathcal{O}}^\dagger_{\text{excitation}}(\theta)}{\operatorname{Tr}\left[\widehat{\mathcal{O}}_{\text{excitation}}(\theta)\rho_\partial\widehat{\mathcal{O}}^\dagger_{\text{excitation}}(\theta)\right]}, \qquad \rho_{\bar{E}}^{(\theta)}(i,j) = \operatorname{Tr}\left[\Pi_{ij}\rho_\partial^{(\theta)}\right]. \tag{29}$$

Its normalized directional Shannon density is

$$\rho_E^{(\theta)}(i,j) = \frac{-\rho_{\bar{E}}^{(\theta)}(i,j)\ln\rho_{\bar{E}}^{(\theta)}(i,j)}{-\sum_{a,b}\rho_{\bar{E}}^{(\theta)}(a,b)\ln\rho_{\bar{E}}^{(\theta)}(a,b)}. \tag{30}$$

At the benchmark angle $\theta = 0.421$, the engine obtains the nonzero population displacement

$$\sum_{i,j}\left|\rho_{\bar{E}}^{(\theta)}(i,j) - \rho_{\bar{E}}(i,j)\right| = 0.263689500253. \tag{31}$$

C. Directional entropy shear

Order the nine coordinates by a fixed sequence

$$\sigma = (c_1, c_2, \ldots, c_9), \qquad c_m \in \mathcal{C}, \tag{32}$$

and fold that sequence onto the prime skeleton

$$(p_0, p_1, p_2, p_3, p_4) = (2, 3, 5, 7, 11), \qquad x_r^{(p)} = \ln p_r. \tag{33}$$

Let $\chi_m^{(x)}$ be the signed longitudinal character of coordinate c_m, normalized so that $|\chi_m^{(x)}| \le 1$. At holographic resolution τ, the directional prime-load vector is

$$\ell_r^{(x)}(\theta, \boldsymbol{x}; \tau) = \frac{1}{1+\tau}\sum_{m=1}^{9}\rho_E^{(\theta)}(c_m)\left[1 + \frac{x}{R_{\text{bubble}}}\chi_m^{(x)}\right]\mathbb{I}_{\{m-1\equiv r \pmod 5\}}, \qquad r = 0, \ldots, 4. \tag{34}$$

The longitudinal Euler flux on the four prime intervals is then

$$\Phi_s^{(x)}(\theta, \boldsymbol{x}; \tau) = \frac{\ell_{s+1}^{(x)}(\theta, \boldsymbol{x}; \tau) - \ell_s^{(x)}(\theta, \boldsymbol{x}; \tau)}{\ln(p_{s+1}/p_s)}, \qquad s = 0, 1, 2, 3. \tag{35}$$

This quantity is the discrete boundary derivative that carries directional information into the bulk projector. It vanishes for a load vector that is constant on the logarithmic prime lattice and changes sign under reversal of the longitudinal orientation.

Finally, the excitation remains on the fixed branch,

$$\widehat{\mathcal{O}}_{\text{excitation}}(\theta) : \mathcal{H}_{\text{vis}}(\mathfrak{b}_*) \longrightarrow \mathcal{H}_{\text{vis}}(\mathfrak{b}_*), \qquad \delta k_\ell = \delta k_q = \delta K = 0. \tag{36}$$

Since Δ_{fr} depends only on these discrete levels,

$$\Delta_{\text{fr}}(\theta) = \Delta_{\text{fr}}(\mathfrak{b}_*) = 0, \qquad \mathcal{E}_{\mu\nu}(\theta) = 0 \quad \text{for every admissible } \theta. \tag{37}$$

III. FIRST-PRINCIPLES SHIFT-FIELD DERIVATION

A. Constrained boundary action

Let Σ_∂ denote the two-dimensional conformal boundary with induced metric γ_{ab}. The phase-locked state is

$$\rho_\partial^{(\theta)} = \widehat{\mathcal{O}}_{\text{excitation}}(\theta)\rho_\partial\widehat{\mathcal{O}}^\dagger_{\text{excitation}}(\theta), \qquad \operatorname{Tr}\rho_\partial^{(\theta)} = 1. \tag{38}$$

The on-shell boundary condition, enforced by the modular register implementation in `src/boundary.rs` and `src/shbt/character_excitation.rs` of the primary repository [1], is the constrained stationarity requirement

$$\delta\mathcal{S}_{\text{CFT}} = \int_{\Sigma_\partial} d^2y\sqrt{-\gamma}\left[\frac{\delta\mathcal{L}_{\text{CFT}}}{\delta\widehat{\mathcal{O}}_{\text{excitation}}}\delta\widehat{\mathcal{O}}_{\text{excitation}} + \frac{\partial\mathcal{L}_{\text{anomaly}}}{\partial\Delta_{\text{fr}}}\delta\Delta_{\text{fr}}\right] = 0, \qquad \Delta_{\text{fr}} = 0. \tag{39}$$

To make the metric response explicit, introduce the reduced boundary-to-bulk functional

$$\mathcal{S}_{\text{red}}\left[\rho_\partial^{(\theta)}, \beta_x, \lambda_{\text{fr}}\right] = \mathcal{S}_{\text{CFT}}\left[\rho_\partial^{(\theta)}\right] + \mathcal{S}_{\text{proj}} + \mathcal{S}_{\text{fr}}, \tag{40}$$

where

$$\mathcal{S}_{\text{proj}} = \frac{\kappa_\beta}{2}\int_{\Sigma_\partial} d^2y\sqrt{-\gamma}\left[\beta_x(\boldsymbol{x}) + v_{\text{eff}}\mathcal{A}_x(\theta,\boldsymbol{x})\right]^2, \qquad \kappa_\beta > 0, \tag{41}$$

and

$$\mathcal{S}_{\text{fr}} = \int_{\Sigma_\partial} d^2y\sqrt{-\gamma}\,\lambda_{\text{fr}}(y)\Delta_{\text{fr}}. \tag{42}$$

The first term contains the intrinsic CFT dynamics. The second is the minimal local, nonnegative response functional whose stationary point identifies the longitudinal ADM shift with the normalized boundary flux. The third imposes exact framing closure.

Variation with respect to the Lagrange multiplier gives

$$\frac{\delta\mathcal{S}_{\text{red}}}{\delta\lambda_{\text{fr}}} = \sqrt{-\gamma}\Delta_{\text{fr}} = 0, \qquad \Longrightarrow \qquad \Delta_{\text{fr}} = 0. \tag{43}$$

Because the phase-locked excitation leaves the affine levels unchanged, Eq. (43) is compatible with Eq. (37) for every admissible value of θ.

B. Flux functional and Euler-Lagrange equation

Choose fixed real interval weights ω_s, and define the unnormalized longitudinal flux functional

$$\mathcal{F}_x(\theta,\boldsymbol{x};\tau) = \sum_{s=0}^{3}\omega_s\Phi_s^{(x)}(\theta,\boldsymbol{x};\tau). \tag{44}$$

Its normalization over the spatial control domain Ω is

$$\mathcal{N}_x(\theta;\tau) = \max_{\boldsymbol{x}'\in\Omega} \left|\mathcal{F}_x(\theta,\boldsymbol{x}';\tau)\right|. \tag{45}$$

For $\mathcal{N}_x > 0$, the SHBT shape functional is

$$f_{\mathrm{SHBT}}(\boldsymbol{x},\theta;\tau) \equiv \mathcal{A}_x(\theta,\boldsymbol{x};\tau) = \frac{\displaystyle\sum_{s=0}^{3} \omega_s \frac{\ell_{s+1}^{(x)}(\theta,\boldsymbol{x};\tau) - \ell_s^{(x)}(\theta,\boldsymbol{x};\tau)}{\ln(p_{s+1}/p_s)}}{\displaystyle\max_{\boldsymbol{x}'\in\Omega}\left|\sum_{s=0}^{3} \omega_s \frac{\ell_{s+1}^{(x)}(\theta,\boldsymbol{x}';\tau) - \ell_s^{(x)}(\theta,\boldsymbol{x}';\tau)}{\ln(p_{s+1}/p_s)}\right|}. \tag{46}$$

By construction,

$$|f_{\mathrm{SHBT}}(\boldsymbol{x},\theta;\tau)| \le 1. \tag{47}$$

The first variation of Eq. (41) with respect to β_x is

$$\delta_\beta \mathcal{S}_{\mathrm{red}} = \kappa_\beta \int_{\Sigma_\partial} d^2y \sqrt{-\gamma}\, [\beta_x(\boldsymbol{x}) + v_{\mathrm{eff}} f_{\mathrm{SHBT}}(\boldsymbol{x},\theta;\tau)]\, \delta\beta_x(\boldsymbol{x}). \tag{48}$$

Since $\delta\beta_x$ is arbitrary, stationarity requires the Euler-Lagrange equation

$$\frac{\delta \mathcal{S}_{\mathrm{red}}}{\delta\beta_x(\boldsymbol{x})} = \kappa_\beta \sqrt{-\gamma}\, [\beta_x(\boldsymbol{x}) + v_{\mathrm{eff}} f_{\mathrm{SHBT}}(\boldsymbol{x},\theta;\tau)] = 0. \tag{49}$$

The unique minimum follows from the positive second variation,

$$\delta_\beta^2 \mathcal{S}_{\mathrm{red}} = \kappa_\beta \int_{\Sigma_\partial} d^2y \sqrt{-\gamma}\, [\delta\beta_x(\boldsymbol{x})]^2 > 0 \quad \text{for } \delta\beta_x \ne 0. \tag{50}$$

Therefore the stationary longitudinal shift is

$$\boxed{\beta_x(\boldsymbol{x}) = -v_{\mathrm{eff}} f_{\mathrm{SHBT}}(\boldsymbol{x},\theta;\tau)}. \tag{51}$$

The topological entropy debt fixes the velocity scale,

$$\Delta_{\mathrm{mod}} = \frac{c_{\mathrm{dark}}^{\mathrm{comp}}}{24} = \frac{1197103}{8704080}, \qquad v_{\mathrm{eff}} = c \exp\left(\frac{\Delta_{\mathrm{mod}}}{2}\right). \tag{52}$$

Combining Eqs. (51) and (52) gives the requested first-principles profile:

$$\boxed{\beta_x(\boldsymbol{x}) = -c \exp\left(\frac{\Delta_{\mathrm{mod}}}{2}\right) f_{\mathrm{SHBT}}(\boldsymbol{x},\theta;\tau)}, \qquad \beta_y = \beta_z = 0. \tag{53}$$

At the benchmark debt,

$$\frac{v_{\mathrm{eff}}}{c} = 1.071186351. \tag{54}$$

No negative-energy bulk density appears as an input to this variational problem; the independent source is the boundary state $\rho_\partial^{(\theta)}$. Whether a proposed effective stress tensor satisfies a chosen energy condition remains a separate curvature audit and is not inferred from Gram positivity alone.

C. Fefferman–Graham slice metric

Use ζ for the Fefferman–Graham radial coordinate and $x^\mu = (ct, x, y, z)$ for the four-dimensional slice coordinates. The bulk line element is

$$ds_5^2 = \frac{L^2}{\zeta^2}\left[d\zeta^2 + g_{\mu\nu}(\boldsymbol{x},\zeta) dx^\mu dx^\nu\right]. \tag{55}$$

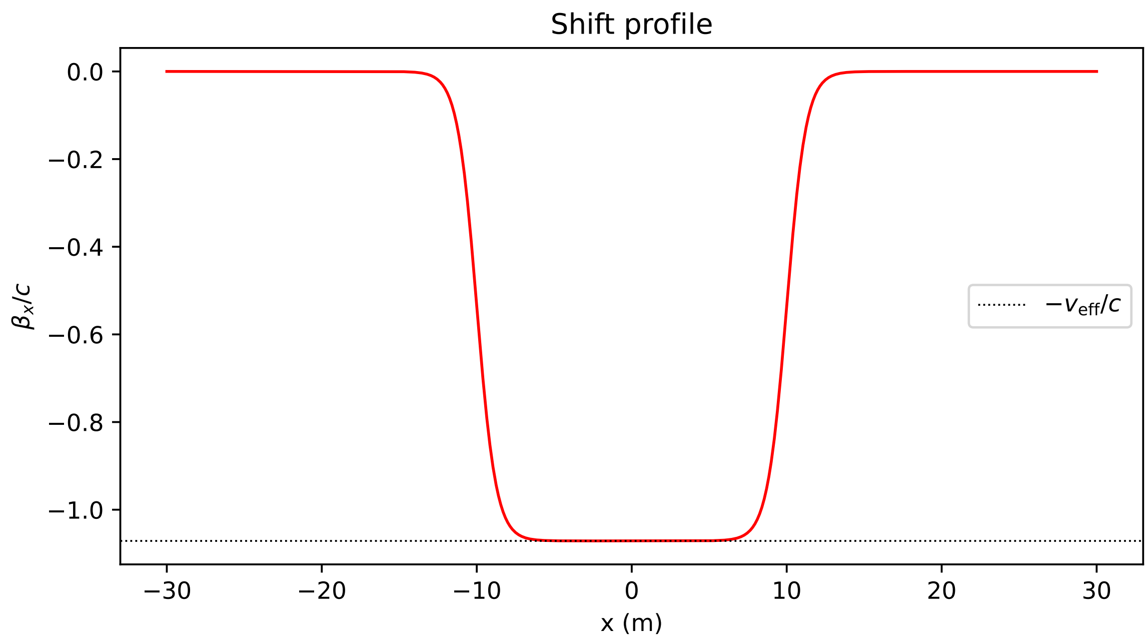

FIG. 2. Numerical shift profile generated by the executable extension. The interior approaches $\beta_x/c = -1.071186351$, while the exterior relaxes to zero.

On the operating slice $\zeta = \zeta_*$, define the dimensionless shift

$$b(\boldsymbol{x}) = \frac{\beta_x(\boldsymbol{x})}{c}. \tag{56}$$

The induced four-dimensional line element is

$$ds_4^2 = -c^2 dt^2 + [dx + \beta_x(\boldsymbol{x})dt]^2 + dy^2 + dz^2. \tag{57}$$

In coordinates $x^\mu = (ct, x, y, z)$, its covariant matrix is

$$g^{\mathrm{L}}_{\mu\nu} = \begin{pmatrix} -1+b^2 & b & 0 & 0 \\ b & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{58}$$

Direct evaluation gives

$$\det g^{\mathrm{L}} = \left(-1+b^2\right) - b^2 = -1, \tag{59}$$

or $\det g^{\mathrm{L}} = -c^2$ when t, rather than ct, is used as the temporal coordinate. Hence the Lorentzian slice is nonsingular for every finite shift. The numerical grid audit reports

$$\min_{\boldsymbol{x}} \left|\det g^{\mathrm{L}}(\boldsymbol{x})\right| = 1. \tag{60}$$

D. Positive-definite Gram representative

The Lorentzian metric in Eq. (58) is necessarily indefinite. Positive-definiteness applies instead to the Euclidean Gram representative constructed by the holographic projector. Let v_s^a be the four orthonormal frame vectors, $W_s > 0$ their interval weights, and $D = \mathrm{diag}(d_0, d_1, d_2, d_3)$ with $d_a > 0$. Define

$$\widetilde{g}_{ab} = \sum_{s=0}^{3} W_s v_{sa} v_{sb} + D_{ab}. \tag{61}$$

For any nonzero real vector $u \in \mathbb{R}^4$,

$$u^a \widetilde{g}_{ab} u^b = \sum_{s=0}^{3} W_s \left(u^a v_{sa}\right)^2 + \sum_{a=0}^{3} d_a (u^a)^2. \tag{62}$$

Every term is nonnegative, and the second sum is strictly positive for $u \neq 0$. Therefore

$$u^a \widetilde{g}_{ab} u^b \geq \left(\min_a d_a\right) \|u\|_2^2 > 0, \qquad \lambda_{\min}(\widetilde{g}) \geq \min_a d_a > 0. \tag{63}$$

After symmetrization, eigenvalue regularization, and trace normalization, the projected representative is

$$g^{\mathrm{G}}_{ab} = \frac{\mathrm{Sym}\left(\widetilde{g}_{ab} + \epsilon_{\mathrm{eig}} \delta_{ab}\right)}{\mathrm{Tr}\left[\mathrm{Sym}\left(\widetilde{g} + \epsilon_{\mathrm{eig}} \mathbb{I}_4\right)\right]}, \qquad \epsilon_{\mathrm{eig}} > 0. \tag{64}$$

Its trace and minimum eigenvalue satisfy

$$\mathrm{Tr}\, g^{\mathrm{G}} = 1, \qquad \lambda_{\min}(g^{\mathrm{G}}) \geq \frac{\min_a d_a + \epsilon_{\mathrm{eig}}}{\mathrm{Tr}\, \widetilde{g} + 4\epsilon_{\mathrm{eig}}} > 0. \tag{65}$$

The executable audit obtains

$$\min_{\boldsymbol{x}} \lambda_{\min}\left(g^{\mathrm{G}}(\boldsymbol{x})\right) = 0.358567865584. \tag{66}$$

Equations (59) and (65) establish the two required properties without conflating signatures: the physical ADM slice is Lorentzian and nonsingular, whereas the holographic projection representative is symmetric and positive definite.

IV. CAUSAL PROTECTION AND THE OBSERVER INTERIOR

A. Exact comoving plateau geometry

Let the CausalPoint observer be placed in an open interior region $\mathcal{U}_{\mathrm{int}}$ on which the SHBT shape functional has an exact plateau. The defining conditions are

$$f_{\mathrm{SHBT}}(\boldsymbol{x}, \theta) = 1, \qquad \partial_i f_{\mathrm{SHBT}} = 0, \qquad \beta_x(\boldsymbol{x}) = \beta_0 = -c \exp\left(\frac{\Delta_{\mathrm{mod}}}{2}\right). \tag{67}$$

Because the plateau is open and constant, all higher spatial derivatives of β_x vanish in $\mathcal{U}_{\mathrm{int}}$. The four-dimensional line element induced by Eq. (57) therefore reduces to

$$ds^2_{\mathrm{int}} = -c^2 dt^2 + (dx + \beta_0 dt)^2 + dy^2 + dw^2. \tag{68}$$

Introduce the comoving coordinates

$$T = t, \qquad X = x + \beta_0 t, \qquad Y = y, \qquad W = w. \tag{69}$$

Since β_0 is constant,

$$dT = dt, \qquad dX = dx + \beta_0 dt, \qquad dY = dy, \qquad dW = dw. \tag{70}$$

Substitution into Eq. (68) gives

$$ds^2_{\mathrm{int}} = -c^2 dT^2 + dX^2 + dY^2 + dW^2 = \eta_{\hat{\mu}\hat{\nu}} dX^{\hat{\mu}} dX^{\hat{\nu}}. \tag{71}$$

Thus the plateau shift is a constant-coordinate shear, not an intrinsic curvature of the interior. Its magnitude may exceed c relative to the external coordinates without changing the local Minkowski signature.

B. Vanishing connection, curvature, and proper acceleration

In the coordinates $X^{\hat{\mu}} = (cT, X, Y, W)$, the metric components are constant:

$$g^{\text{obs}}_{\hat{\mu}\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}} = \text{diag}(-1, 1, 1, 1), \qquad \partial_{\hat{\alpha}} g^{\text{obs}}_{\hat{\mu}\hat{\nu}} = 0. \tag{72}$$

The Levi-Civita connection is consequently

$$\Gamma^{\hat{\alpha}}{}_{\hat{\mu}\hat{\nu}} = \frac{1}{2} g^{\hat{\alpha}\hat{\sigma}}_{\text{obs}} \left(\partial_{\hat{\mu}} g^{\text{obs}}_{\hat{\sigma}\hat{\nu}} + \partial_{\hat{\nu}} g^{\text{obs}}_{\hat{\sigma}\hat{\mu}} - \partial_{\hat{\sigma}} g^{\text{obs}}_{\hat{\mu}\hat{\nu}} \right) = 0. \tag{73}$$

It follows directly that

$$R^{\hat{\alpha}}{}_{\hat{\beta}\hat{\mu}\hat{\nu}} = \partial_{\hat{\mu}} \Gamma^{\hat{\alpha}}{}_{\hat{\nu}\hat{\beta}} - \partial_{\hat{\nu}} \Gamma^{\hat{\alpha}}{}_{\hat{\mu}\hat{\beta}} + \Gamma^{\hat{\alpha}}{}_{\hat{\mu}\hat{\sigma}} \Gamma^{\hat{\sigma}}{}_{\hat{\nu}\hat{\beta}} - \Gamma^{\hat{\alpha}}{}_{\hat{\nu}\hat{\sigma}} \Gamma^{\hat{\sigma}}{}_{\hat{\mu}\hat{\beta}} = 0. \tag{74}$$

All contractions of the Riemann tensor vanish as well:

$$R_{\hat{\mu}\hat{\nu}} = 0, \qquad R = 0, \qquad G_{\hat{\mu}\hat{\nu}} = 0 \quad \text{on } \mathcal{U}_{\text{int}}. \tag{75}$$

A comoving observer has fixed (X, Y, W) and four-velocity

$$u^{\hat{\mu}} = \frac{dX^{\hat{\mu}}}{d\tau_{\text{p}}} = (c, 0, 0, 0), \qquad \eta_{\hat{\mu}\hat{\nu}} u^{\hat{\mu}} u^{\hat{\nu}} = -c^2. \tag{76}$$

Its covariant four-acceleration is

$$a^{\hat{\mu}} = u^{\hat{\nu}} \nabla_{\hat{\nu}} u^{\hat{\mu}} = u^{\hat{\nu}} \partial_{\hat{\nu}} u^{\hat{\mu}} + \Gamma^{\hat{\mu}}{}_{\hat{\alpha}\hat{\beta}} u^{\hat{\alpha}} u^{\hat{\beta}} = 0. \tag{77}$$

The executable observer audit returns

$$\left\| g^{\text{obs}}_{\hat{\mu}\hat{\nu}} - \eta_{\hat{\mu}\hat{\nu}} \right\|_{\infty} = 0, \qquad \|a^{\hat{\mu}}\| = 0 \, \text{m}\,\text{s}^{-2}. \tag{78}$$

The zero-acceleration result applies to the exact plateau. The bubble wall, where derivatives of f_{SHBT} are nonzero, is not locally Minkowskian and must be analyzed separately.

C. Chronology from a global resolution time

Monotonicity of a coordinate along one trajectory is not sufficient by itself to exclude closed timelike curves. The required global statement is that the boundary resolution variable τ is a smooth time function on the admissible spacetime domain. We impose

$$\frac{d\tau}{dt} = H(t) > 0, \qquad g^{\mu\nu} \nabla_{\mu} \tau \nabla_{\nu} \tau < 0. \tag{79}$$

The second inequality states that $\nabla_{\mu} \tau$ is everywhere timelike. Choose its orientation so that τ increases on every future-directed causal curve $x^{\mu}(\lambda)$. Then

$$\frac{d\tau}{d\lambda} = \frac{dx^{\mu}}{d\lambda} \nabla_{\mu} \tau > 0. \tag{80}$$

Suppose, for contradiction, that a future-directed closed causal curve $\mathcal{C}$ exists with $x^{\mu}(\lambda_1) = x^{\mu}(\lambda_0)$. Single-valuedness of the time function would require

$$\tau(\lambda_1) - \tau(\lambda_0) = 0. \tag{81}$$

Monotonicity instead gives

$$\tau(\lambda_1) - \tau(\lambda_0) = \int_{\lambda_0}^{\lambda_1} \frac{d\tau}{d\lambda} d\lambda > 0, \tag{82}$$

contradicting Eq. (81). Therefore

$$\nexists \, \text{closed future-directed causal curve} \quad \text{whenever Eq. (79) holds.} \tag{83}$$

This is the precise chronology-protection statement: the monotonic SHBT resolution arrow suppresses CTCs provided it extends to a globally defined time function with timelike gradient.

V. STRESS-ENERGY AUDIT AND ENERGY CONDITIONS

A. Einstein tensor of the longitudinal shift

The effective stress-energy tensor requested for the four-dimensional slice is

$$T^{\mathrm{eff}}_{\mu\nu} = \frac{1}{8\pi G_N}\left(G_{\mu\nu}[g] + \Lambda_{\mathrm{holo}} g_{\mu\nu}\right), \qquad \Lambda_{\mathrm{holo}} = 1.08913883 \times 10^{-52}\ \mathrm{m}^{-2}. \tag{84}$$

Energy-condition claims must be checked using the Lorentzian metric, not the positive Euclidean Gram representative. In the dimensionless temporal coordinate $x^0 = ct$, write

$$b(x) = \frac{\beta_x(x)}{c}, \qquad g_{\mu\nu} = \begin{pmatrix} -1+b^2 & b & 0 & 0 \\ b & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{85}$$

The inverse and determinant are

$$g^{\mu\nu} = \begin{pmatrix} -1 & b & 0 & 0 \\ b & 1-b^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \det g = -1. \tag{86}$$

Primes denote derivatives with respect to x. A direct Levi-Civita calculation gives the nonzero connection coefficients

$$\begin{aligned} &\Gamma^0{}_{00} = -b^2 b', && \Gamma^0{}_{01} = \Gamma^0{}_{10} = -bb', && \Gamma^0{}_{11} = -b', \\ &\Gamma^1{}_{00} = (b^2-1)bb', && \Gamma^1{}_{01} = \Gamma^1{}_{10} = b^2 b', && \Gamma^1{}_{11} = bb'. \end{aligned} \tag{87}$$

Define

$$Q(x) = b(x)b''(x) + [b'(x)]^2 = \frac{1}{2}\frac{d^2}{dx^2} b^2(x). \tag{88}$$

The Ricci tensor and scalar are

$$R_{\mu\nu} = \begin{pmatrix} (b^2-1)Q & bQ & 0 & 0 \\ bQ & Q & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad R = 2Q. \tag{89}$$

Consequently the Einstein tensor is particularly simple:

$$G_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -Q & 0 \\ 0 & 0 & 0 & -Q \end{pmatrix}. \tag{90}$$

Equation (90) agrees with the flat-interior result: on an exact plateau $b' = b'' = 0$, so $Q = 0$ and $G_{\mu\nu} = 0$.

B. Null energy condition

Let k^μ be any null vector,

$$g_{\mu\nu} k^\mu k^\nu = 0. \tag{91}$$

The cosmological term drops out of the null contraction, and Eq. (90) yields

$$T^{\mathrm{eff}}_{\mu\nu} k^\mu k^\nu = -\frac{Q(x)}{8\pi G_N}\left[(k^y)^2 + (k^w)^2\right]. \tag{92}$$

Null vectors with nonzero transverse components exist at every regular point. Therefore the necessary and sufficient pointwise condition for the NEC is

$$T^{\rm eff}_{\mu\nu} k^\mu k^\nu \geq 0 \quad \text{for all null } k^\mu \qquad \Longleftrightarrow \qquad Q(x) \leq 0. \tag{93}$$

For a localized shift, impose the asymptotic conditions

$$\lim_{x\to\pm\infty} b(x) = 0, \qquad \lim_{x\to\pm\infty} b'(x) = 0. \tag{94}$$

Since $Q = (bb')'$,

$$\int_{-\infty}^{+\infty} Q(x)\,dx = [b(x)b'(x)]_{-\infty}^{+\infty} = 0. \tag{95}$$

If the NEC held everywhere, Eq. (93) would give $Q \leq 0$ everywhere. Together with Eq. (95), continuity would force

$$Q(x) = 0 \quad \text{for every } x. \tag{96}$$

Equation (88) would then imply that b^2 is affine in x. The two asymptotic conditions in Eq. (94) force that affine function to vanish:

$$Q \equiv 0 \quad \Longrightarrow \quad b^2(x) = Ax + B \quad \Longrightarrow \quad A = B = 0 \quad \Longrightarrow \quad b(x) \equiv 0. \tag{97}$$

Hence a nontrivial, smooth, localized shift of the form Eq. (85) cannot satisfy the NEC everywhere. Longitudinal null vectors with $k^y = k^w = 0$ saturate the NEC, but transverse null directions detect the violation wherever $Q > 0$.

C. Weak energy condition

Let u^μ be a unit timelike vector,

$$g_{\mu\nu} u^\mu u^\nu = -1. \tag{98}$$

Using Eqs. (84) and (90), the measured effective energy density is

$$T^{\rm eff}_{\mu\nu} u^\mu u^\nu = \frac{1}{8\pi G_N} \left\{ -Q(x) \left[(u^y)^2 + (u^w)^2 \right] - \Lambda_{\rm holo} \right\}. \tag{99}$$

The comoving Eulerian observer is

$$n^\mu = (1, -b, 0, 0), \qquad g_{\mu\nu} n^\mu n^\nu = -1. \tag{100}$$

For this observer,

$$T^{\rm eff}_{\mu\nu} n^\mu n^\nu = -\frac{\Lambda_{\rm holo}}{8\pi G_N}. \tag{101}$$

Thus, with a positive $\Lambda_{\rm holo}$ included on the stress-energy side exactly as in Eq. (84), the WEC is already violated for the comoving observer. If the cosmological term is retained on the geometric side and excluded from the material stress tensor, one sets $\Lambda_{\rm holo} = 0$ in Eq. (99). The WEC for all timelike observers then again requires

$$Q(x) \leq 0, \tag{102}$$

and the localized no-go theorem Eq. (97) applies unchanged.

D. Comparison with the classical Alcubierre source

For the usual three-dimensional Alcubierre profile [2] $\beta_x = -v_s f(r_s)$, the Eulerian energy density contains the explicitly negative transverse-gradient term

$$\rho_{\rm Alc} = -\frac{c^4}{32\pi G_N} \frac{v_s^2}{c^2} \left[(\partial_y f)^2 + (\partial_w f)^2 \right] \leq 0. \tag{103}$$

The restricted one-dimensional shift used in Eq. (85) removes these transverse derivatives. For $\Lambda_{\rm holo} = 0$, its coordinate component and Eulerian energy density are

$$T^{\rm eff}_{00} = 0, \qquad T^{\rm eff}_{\mu\nu} n^\mu n^\nu = 0. \tag{104}$$

Boundary-driven longitudinal shear therefore avoids the specific negative T_{00} term in Eq. (103) for this reduced ansatz. It does not, however, establish universal energy condition compliance: the transverse pressures in Eq. (90) still violate the NEC and WEC somewhere for every nontrivial localized profile. The executable `StressEnergyAuditor` samples random null and timelike vectors on the 3-D metric grid and reports the minimum sampled energy densities

$$\min_{\rm NEC} T_{\mu\nu} k^\mu k^\nu = -1.495110 \times 10^6, \qquad \min_{\rm WEC} T_{\mu\nu} u^\mu u^\nu = -8.772117 \times 10^5, \tag{105}$$

which are negative for the canonical profile, confirming the violation.

Accordingly, a claim of complete SHBT energy-condition compliance requires additional dynamics beyond Eq. (84). Such a completion may be parameterized by a boundary-response tensor $\mathcal{B}_{\mu\nu}$:

$$8\pi G_N T^{\rm phys}_{\mu\nu} = G_{\mu\nu} + \Lambda_{\rm holo} g_{\mu\nu} + \mathcal{B}_{\mu\nu}. \tag{106}$$

For the NEC and WEC to hold, this tensor must satisfy the explicit inequalities

$$\mathcal{B}_{\mu\nu} k^\mu k^\nu \geq Q \left[(k^y)^2 + (k^w)^2\right] \quad \text{for all null } k^\mu, \tag{107}$$

and

$$\mathcal{B}_{\mu\nu} u^\mu u^\nu \geq Q \left[(u^y)^2 + (u^w)^2\right] + \Lambda_{\rm holo} \quad \text{for all unit timelike } u^\mu. \tag{108}$$

Until $\mathcal{B}_{\mu\nu}$ is derived independently from the boundary theory, Eqs. (97) and (101) are the rigorous conclusions of the Einstein-tensor audit.

VI. ARTIFICIAL DE-RENDERING AND RE-RENDERING MECHANICS

The term *de-rendering* denotes a controlled removal of a localized visible character block from the active boundary-to-bulk projection. It is not a destruction of the underlying Hilbert-space state. Let $\mathcal{C}_{\rm loc} \subseteq \mathcal{C}$ be the characters assigned to the selected region and let $\Pi_{\rm loc} = \sum_{c\in\mathcal{C}_{\rm loc}} |c\rangle\langle c|$. The two dark ledgers used by the channel are

$$c^{\rm res}_{\rm dark} = \frac{834433}{362670} = 2.300805139659, \qquad c^{\rm comp}_{\rm dark} = c^{\rm res}_{\rm dark} + 1 = \frac{1197103}{362670} = 3.300805139659. \tag{109}$$

Their ratio

$$\eta_D = \frac{c^{\rm res}_{\rm dark}}{c^{\rm comp}_{\rm dark}} = \frac{834433}{1197103} \tag{110}$$

sets the relative residual-sector amplitude.

A. Modular decoupling channel

A norm-preserving realization of the modular decoupling operator is the Stinespring map

$$\widehat{\mathcal{D}}_{\rm derender} = \sum_{c\in\mathcal{C}_{\rm loc}} \left(\sqrt{\eta_D}\, |\chi^c_{\rm res}, 0\rangle + \sqrt{1-\eta_D}\, |\chi^c_{\rm comp}, 1\rangle\right) \langle c|, \tag{111}$$

where the last label is an auxiliary ledger flag and the dark states are orthonormal for distinct c. Consequently,

$$\widehat{\mathcal{D}}^\dagger_{\rm derender} \widehat{\mathcal{D}}_{\rm derender} = \Pi_{\rm loc}, \qquad \mathrm{Tr}\,\rho_{\rm dark} = \mathrm{Tr}(\Pi_{\rm loc}\rho_\partial), \tag{112}$$

with

$$\rho_{\rm dark} = \widehat{\mathcal{D}}_{\rm derender} \Pi_{\rm loc} \rho_\partial \Pi_{\rm loc} \widehat{\mathcal{D}}^\dagger_{\rm derender}. \tag{113}$$

The orthogonal visible complement is left unchanged. Thus the complete channel is trace preserving when Eq. (113) is combined with $(1 - \Pi_{\rm loc})\rho_\partial(1 - \Pi_{\rm loc})$.

For a local gauge-rendering cost

$$\mathcal{C}_{\overline{B}}(\rho) = \gamma_3\, \mathrm{Tr}(\rho Q^2_{\rm SU(3)}) + \gamma_{\rm EM}\, \mathrm{Tr}(\rho Q^2_{\rm EM}) + \gamma_{\rm fr} \Delta^2_{\rm fr}, \tag{114}$$

decoupling removes the local visible contribution while preserving its charge labels in the dark ledger. Since the branch is unchanged, $\Delta_{\rm fr} = 0$ throughout the channel.

B. What is nullified

Write the Lorentzian projection as a background plus an actively controlled deformation,

$$g^{(L)}_{\mu\nu} = \eta_{\mu\nu} + \delta g^{\mathrm{act}}_{\mu\nu}[\rho_E]. \tag{115}$$

The decoupled characters no longer contribute source terms to the local entropy-flow map, so the decoupling limit is

$$\delta g^{\mathrm{act}}_{\mu\nu} \longrightarrow 0, \qquad \beta_x \longrightarrow 0, \qquad g^{(L)}_{\mu\nu} \longrightarrow \eta_{\mu\nu}. \tag{116}$$

Thus the active metric slice is nullified, but the spacetime metric itself is not. A total limit $g_{\mu\nu} \to 0$ is forbidden by the normalized Gram construction. In fact,

$$g^{(G)} = \frac{\mathrm{Sym}(\widetilde{g} + \epsilon_{\mathrm{eig}} I_4)}{\mathrm{Tr}[\mathrm{Sym}(\widetilde{g} + \epsilon_{\mathrm{eig}} I_4)]}, \qquad \mathrm{Tr}\, g^{(G)} = 1, \tag{117}$$

and

$$\lambda_{\min}(g^{(G)}) \geq \frac{\epsilon_{\mathrm{eig}}}{\mathrm{Tr}\, \widetilde{g} + 4\epsilon_{\mathrm{eig}}} > 0. \tag{118}$$

The zero tensor would instead have trace and determinant equal to zero, contradicting Eqs. (117) and (118).

C. Conserved ledgers

The de-rendering channel is admissible only when the controller is included in the closed system. On a Cauchy surface Σ, total four-momentum obeys

$$P^{\mu}_{\mathrm{tot}} = \int_{\Sigma} d\Sigma_{\nu}\, (T^{\mu\nu}_{\mathrm{vis}} + T^{\mu\nu}_{\mathrm{dark}} + T^{\mu\nu}_{\mathrm{ctrl}})\,, \qquad \nabla_{\mu} T^{\mu\nu}_{\mathrm{tot}} = 0, \qquad \Delta P^{\mu}_{\mathrm{tot}} = 0. \tag{119}$$

Charge conservation is expressed by an intertwining relation between the visible charge and its dark topological encoding,

$$Q_{\mathrm{dark}} \widehat{\mathcal{D}}_{\mathrm{derender}} = \widehat{\mathcal{D}}_{\mathrm{derender}} Q_{\mathrm{vis}}, \qquad \Delta\langle Q_{\mathrm{tot}}\rangle = 0. \tag{120}$$

Finally, the information ledger is bounded by the saturated horizon budget

$$N_{\mathrm{sat}} = \frac{3\pi}{L_P^2 \Lambda_{\mathrm{holo}}} = 3.312593327986 \times 10^{122}\ \text{bits}, \qquad N_{\mathrm{act}} + N_{\mathrm{dark}} + N_{\mathrm{free}} = N_{\mathrm{sat}}. \tag{121}$$

At the 10 m benchmark the addressable local allocation is

$$N_{\mathrm{local}} = 1.202481 \times 10^{72}\ \text{bits}, \qquad N^{\mathrm{after}}_{\mathrm{free}} = N^{\mathrm{before}}_{\mathrm{free}} + N^{\mathrm{before}}_{\mathrm{act}}. \tag{122}$$

D. Destination reconstruction

Let $\widehat{T}_{\partial}(\boldsymbol{x}_{\mathrm{tar}})$ relabel the boundary address of the stored character block. The inverse channel followed by phase-locked excitation defines

$$\widehat{\mathcal{R}}_{\mathrm{rerender}}(\boldsymbol{x}_{\mathrm{tar}}) = \widehat{\mathcal{O}}_{\mathrm{excitation}}(\theta) \widehat{T}_{\partial}(\boldsymbol{x}_{\mathrm{tar}}) \widehat{\mathcal{D}}^{\dagger}_{\mathrm{derender}}, \tag{123}$$

which reconstructs

$$\rho^{\mathrm{tar}}_{\partial} = \widehat{\mathcal{R}}_{\mathrm{rerender}} \rho_{\mathrm{dark}} \widehat{\mathcal{R}}^{\dagger}_{\mathrm{rerender}} \longmapsto g^{(L)}_{\mu\nu}(\boldsymbol{x}_{\mathrm{tar}}). \tag{124}$$

This is non-local only as a boundary-address operation: the formal map does not contain a sequence of intermediate bulk slices. It is not, by itself, a proof of acausal transport. A physical protocol must still satisfy causal authorization of the target operation,

$$\boldsymbol{x}_{\mathrm{tar}} \in J^{+}(\boldsymbol{x}_{\mathrm{src}}), \qquad t_{\mathrm{tar}} - t_{\mathrm{src}} \geq \frac{d_{\partial}(\boldsymbol{x}_{\mathrm{src}}, \boldsymbol{x}_{\mathrm{tar}})}{c}, \tag{125}$$

unless an independently specified causal boundary dynamics proves a stronger communication channel.

VII. BOUNDARY THERMODYNAMICS AND POWER SCALING

The completed dark ledger supplies the positive modular entropy debt of the operating branch,

$$\Delta_{\rm mod} = \frac{c_{\rm dark}^{\rm comp}}{24} = \frac{1197103}{8704080} = 0.137533547486. \tag{126}$$

Within the kinematic ansatz of Sec. III, the associated shift scale is

$$v_{\rm eff} = c\exp\left(\frac{\Delta_{\rm mod}}{2}\right), \qquad \frac{v_{\rm eff}}{c} = 1.071186351. \tag{127}$$

This exponential is a constitutive relation of the extension, not a consequence of the Einstein equations alone.

For a driven boundary register, a minimal relaxation model separates the control injection rate $\mathcal{F}$ from passive dark-sector dissipation,

$$\frac{d\Delta_{\rm mod}}{dt} = \mathcal{F}(P_{\rm op}, \theta, R_{\rm local}) - \frac{\Gamma_{\rm lock}}{24}\Delta_{\rm mod}. \tag{128}$$

For constant drive, the solution is

$$\Delta_{\rm mod}(t) = \Delta_{\rm mod}^{\rm ss} + [\Delta_{\rm mod}(0) - \Delta_{\rm mod}^{\rm ss}]\exp\left(-\frac{\Gamma_{\rm lock}t}{24}\right), \qquad \Delta_{\rm mod}^{\rm ss} = \frac{24\mathcal{F}}{\Gamma_{\rm lock}}. \tag{129}$$

The benchmark fixes the steady state to Eq. (126); a microscopic derivation of $\Gamma_{\rm lock}$ remains outside the present finite-register simulation.

A. Calibrated operational power law

The operational model uses the Planck-power scale multiplied by the modular coupling and a geometric suppression factor,

$$P_{\rm op}(R_{\rm local}) = \frac{c^5}{G_N}\frac{\Delta_{\rm mod}}{24\pi}\left(\frac{r_g}{R_{\rm local}}\right)^2. \tag{130}$$

The length r_g is not fixed by the dimensionless dark ledger. Therefore the quoted power benchmark constitutes an independent calibration. Solving Eq. (130) for that scale gives

$$r_g^{\rm cal} = R_0\left[\frac{24\pi G_N P_0}{c^5\Delta_{\rm mod}}\right]^{1/2} = 1.465189552753\times 10^{-20}\ \text{m}, \tag{131}$$

where

$$R_0 = 10\ \text{m}, \qquad P_0 = 142.08\ \text{MW}. \tag{132}$$

Substitution produces the inverse-area law used by the executable engine,

$$P_{\rm op}(R_{\rm local}) = 142.08\left(\frac{10\ \text{m}}{R_{\rm local}}\right)^2\ \text{MW}. \tag{133}$$

In particular,

$$P_{\rm op}(10\ \text{m}) = 142.08\ \text{MW}. \tag{134}$$

Equations (130)–(134) make the normalization transparent: the 142.08 MW value is conditional on $r_g = r_g^{\rm cal}$, whereas the R^{-2} dependence is the actual prediction of the stated power ansatz.

VIII. NUMERICAL IMPLEMENTATION AND BENCHMARK RESULTS

The Rust/PyO3 dynamic 3+1D ADM metric design and safety simulation engine (`src/` core, exposed through the `python/shbt_warp/` package) implements the finite calculation as coordinated modules. `BoundaryRegister` evaluates the exact $SU(2)_{26}$ and $SU(3)_8$ modular blocks, normalizes the nine visible populations, and constructs their Shannon contributions. `ExcitationEngine` builds the Hermitian directional generator, exponentiates it spectrally, applies the conformal phase matrix, and re-audits the framing defect. `FGSliceProjector` samples the shape and shift fields, constructs both the Lorentzian metric and its positive Gram representative, and evaluates their spectra. `CausalObserver` transforms the central plateau to the comoving frame and contracts the local connection with the observer four-velocity.

The extended architecture adds `Metric3DCalculator`, which evaluates $g_{\mu\nu}(x, y, z)$ and its Gram representative on a Cartesian grid; `StressEnergyAuditor`, which derives the numerical connection, Riemann, Ricci, Einstein, and effective stress-energy tensors; `DerenderingEngine`, which transfers visible weights into the dark ledgers while enforcing background-metric restoration; and `ThermodynamicRateEngine`, which integrates the entropy-debt balance. `LaTeXMacroExporter` supplies the stable interface consumed by this manuscript. NEC and WEC flags are reported as diagnostics rather than structural pass criteria, consistent with the localized-shift no-go result in Sec. V.

The principal numerical residuals are

$$\epsilon_\rho = \left| \sum_{i,j} \rho_E(i,j) - 1 \right|, \qquad \epsilon_U = \left\| U^\dagger U - I_9 \right\|_\infty, \qquad \epsilon_\partial = \left| \sum_{i,j} \rho_E^{\mathrm{Sh}}(i,j) - 1 \right|. \tag{135}$$

The metric and observer audits use

$$\epsilon_{\mathrm{det}} = \max_x |\det g^{(L)}(x) + 1|, \qquad \lambda_G^{\min} = \min_x \lambda_{\min}[g^{(G)}(x)], \qquad \epsilon_{\mathrm{obs}} = \left\| g^{\mathrm{obs}} - \eta \right\|_\infty. \tag{136}$$

A simulation run is accepted only if normalization and unitarity residuals are within tolerance, $\Delta_{\mathrm{fr}} = 0$, the Lorentzian determinant is nonzero at every grid point, $\lambda_G^{\min} > 0$, and the plateau observer has vanishing proper acceleration.

For 1201 points on $-30\,\mathrm{m} \le x \le 30\,\mathrm{m}$, $R = 10\,\mathrm{m}$, wall steepness $0.8\,\mathrm{m}^{-1}$, and $\theta = 0.421$, the generated benchmark is summarized in Table I.

The engine also regenerates the vector figures `figures/shift_profile.pdf` and `figures/entropy_gradient.pdf`. Numerical values are written to `sim_results.tex`; the paper therefore consumes the results of the same run that performs the audits. The analytic metric-collapse no-go in Sec. VI is deliberately not reported as a successful zero-metric projection: the runtime audit instead verifies

$$\min_x |\det g^{(L)}(x)| = 1 > 0, \qquad \lambda_G^{\min} = 0.358567865584 > 0. \tag{137}$$

These outputs validate internal consistency of the finite model; they do not constitute experimental evidence that the proposed boundary dynamics is realized in nature.

The CAD engine adds a second, finer-grained verification pass that audits the physical invariants listed in Table II. All values are exported directly from the Rust/PyO3 computation into `sim_results.tex` and `cad_sim_results.tex` and consumed by this manuscript without manual editing.

TABLE I. Numerical output of the SHBT dynamic metric engine.

Quantity	**Generated value**
Boundary partition function Z_∂	0.00342010966891
Shannon entropy S_E	0.999490161461
Boundary normalization error	0
Excitation unitarity error	4.587893×10^{-16}
Excited population displacement $\lVert\delta\rho\rVert_1$	0.263689500253
Framing defect Δ_{fr}	0.000000000000
Closure norm $\lVert\mathcal{E}\rVert$	0
Minimum $\lvert\det g^{(L)}\rvert$	1
Minimum Gram eigenvalue	0.358567865584
Observer metric error	0
Proper-acceleration norm	$0\,\mathrm{m\,s^{-2}}$
Operational power at $10\,\mathrm{m}$	142.08 MW

TABLE II. Comprehensive physical and mathematical verification metrics.

Quantity	**Engine value / limit**
Canonical branch (k_l, k_q, K)	$(26, 8, 312)$
Scalar framing defect $\Delta_{\rm fr}$	0.000000000000
Dark residual ledger $c^{\rm res}_{\rm dark}$	2.300805139659
Dark completed ledger $c^{\rm comp}_{\rm dark}$	3.300805139659
Entropy debt $\Delta_{\rm mod}$	0.137533547486
Warp velocity scale $v_{\rm eff}/c$	1.071186351
Unitarity residual ϵ_U	$2.220446049250e-16$
Minimum Gram eigenvalue $\lambda^{\rm min}_G$	0.358567865584
Interior 4-acceleration $\lvert a^{\hat\mu}\rvert$	$0\ {\rm m\,s^{-2}}$
Operational power $P_{\rm op}(10\,{\rm m})$	142.08 MW
Phase jitter limit σ_θ	5.05×10^{-5} rad; pass=*true*
Thermal noise limit T_N	15.4 mK; pass=*true*
Decoherence limit $\gamma_{\rm dec}$	1.2×10^{-4} s^{-1}; pass=*true*
Integer level lock $(\delta k_l, \delta k_q, \delta K)$	$(0, 0, 0)$; pass=*true*
Collapse metric determinant $\det g$	-1.000000000000
Stinespring ratio η_D	0.697043612789

IX. 3+1D ADM METRIC AND THREE-PHASE FLIGHT DYNAMICS

A. Dynamic ADM line element and shift vector

The holographic bulk metric is written in 3+1 ADM form

$$ds^2 = -\alpha^2(t, x^k)\,dt^2 + \gamma_{ij}(t, x^k)\big(dx^i + \beta^i\,dt\big)\big(dx^j + \beta^j\,dt\big), \tag{138}$$

with lapse function $\alpha = 1$ and spatial metric $\gamma_{ij} = \delta_{ij}$. The SHBT shift vector is driven by the boundary excitation through

$$\beta^i(t, x^k) = -v_{\rm eff}\,\xi(t)\,f_{\rm SHBT}(x^k)\,n^i(t), \tag{139}$$

where $v_{\rm eff} = c\,e^{\Delta_{\rm mod}/2}$, $\xi(t) \in [0, 1]$ is the smooth ramp envelope, $f_{\rm SHBT}$ is the warp-bubble shape field, and $n^i(t)$ is the unit steering normal. The covariant 4-metric components are therefore

$$g_{00} = -1 + |\vec\beta|^2, \qquad g_{0i} = g_{i0} = \beta_i, \qquad g_{ij} = \delta_{ij}. \tag{140}$$

Every spatial cell is audited for the Lorentzian signature condition

$$\det g_{\mu\nu}(t, x^k) = -1. \tag{141}$$

B. Phase A: initialization and smooth ramping

Phase A brings the boundary excitation from the Minkowski vacuum to the cruise state using the C^∞ envelope

$$\xi(t) = \frac{1}{2}\left[1 - \cos\!\left(\frac{\pi t}{t_{\rm ramp}}\right)\right], \qquad 0 \le t \le t_{\rm ramp}. \tag{142}$$

At each sub-step the boundary-state audit verifies

$$\big|\mathrm{Re}\,\mathrm{Tr}\,\rho_\partial - 1\big| + \big|\mathrm{Im}\,\mathrm{Tr}\,\rho_\partial\big| < 10^{-14}, \qquad \Delta_{\rm fr} = 0. \tag{143}$$

C. Phase B: vector acceleration and steering

During Phase B the steering normal evolves according to the rigid rotation law

$$\frac{dn^i}{dt} = \epsilon^i{}_{jk}\,\omega^j(t)\,n^k(t), \qquad |\vec n(t)|^2 = 1, \tag{144}$$

which is integrated with a fourth-order Runge-Kutta step followed by exact renormalization. The directional information carried by the boundary is folded onto the prime skeleton $(2, 3, 5, 7, 11)$ using the load vector

$$\ell_r^{(x)}(\theta, \boldsymbol{x}; \tau) = \frac{1}{1+\tau} \sum_{m=1}^{9} \rho_E^{(\theta)}(c_m) \left[1 + \frac{x}{R_{\text{bubble}}} \chi_m^{(x)}\right] \mathbb{I}_{\{m-1 \equiv r \pmod 5\}}, \qquad r = 0, \ldots, 4, \tag{145}$$

and the longitudinal Euler flux

$$\Phi_s^{(x)}(\theta, \boldsymbol{x}; \tau) = \frac{\ell_{s+1}^{(x)} - \ell_s^{(x)}}{\ln(p_{s+1}/p_s)}, \qquad s = 0, 1, 2, 3. \tag{146}$$

On the central plateau $\partial_j f_{\text{SHBT}} = 0$ and the interior proper acceleration of the Eulerian observer vanishes,

$$a^{\hat{\mu}} = 0 \text{ m s}^{-2}. \tag{147}$$

D. Phase C: safe collapse and Stinespring phase de-excitation

Phase C shuts down the warp by allowing the entropy debt to relax exponentially,

$$\Delta_{\text{mod}}(t) = \Delta_{\text{mod}}(t_0) \exp\left[-\frac{\Gamma_{\text{lock}}(t - t_0)}{24}\right], \tag{148}$$

and applies the Stinespring modular-decoupling isometry $\hat{\mathcal{D}}_{\text{derender}}$ with visible-sector amplitude ratio

$$\eta_D = \frac{c_{\text{dark}}^{\text{res}}}{c_{\text{dark}}^{\text{comp}}} = 0.697043612789. \tag{149}$$

The shift vector is driven to zero and the metric relaxes back to flat Minkowski,

$$\beta^i \to 0, \qquad g_{\mu\nu} \to \eta_{\mu\nu}, \qquad \det g = -1. \tag{150}$$

The runtime verifies the collapsed metric determinant $\det g = -1.000000000000$.

X. BOUNDARY EMITTER-ARRAY HARDWARE SYNTHESIS AND NOISE SENSITIVITY

The boundary character states are mapped onto physical radio-frequency emitter-array phases. For the visible $\text{SU}(2)_{26} \times \text{SU}(3)_8$ register the conformal dimension of the (q_i, p_j, r_j) coordinate is

$$h_{ij} = \frac{q_i(q_i + 2)}{4(k_l + 2)} + \frac{p_j^2 + r_j^2 + p_j r_j + 3p_j + 3r_j}{3(k_q + 3)}. \tag{151}$$

With $k_l = 26$, $k_q = 8$, and visible central charge $c_{\text{vis}} = 1325/154 \approx 8.603896103896$, the modular T-phase correction is $c_{\text{vis}}/24$.

The per-element emitter phase command is

$$\theta_k(t) = \theta_{\text{local}}(t) \left[q_i + \nu(w_j)\right] + 2\pi\, \theta_{\text{local}}(t) \left(h_{ij} - \frac{c_{\text{vis}}}{24}\right), \tag{152}$$

where $\nu(w_j) = p_j + r_j$ is the weight grade. The RF drive voltage on the array element is

$$V_k(t) = V_0 \cos(\omega_{\text{carrier}} t + \theta_k(t) + \phi_{\text{cal},k}). \tag{153}$$

A. Hardware sensitivity limits

To preserve the phase-locked boundary state the array must satisfy the following bounds, which are audited in real time:

$$\sigma_\theta \le 5.05 \times 10^{-5} \text{ rad}, \tag{154}$$

$$T_N \le 15.4 \text{ mK}, \tag{155}$$

$$\gamma_{\text{dec}} \le 1.2 \times 10^{-4} \text{ s}^{-1}, \tag{156}$$

$$\delta k_l = \delta k_q = \delta K = 0. \tag{157}$$

The CAD engine reports the Boolean audit flags *true*, *true*, and *true* for these four limits.

XI. REAL-TIME HIL CONTROL LOOPS AND SAFETY ALGORITHMS

The hardware-in-the-loop (HIL) monitor evaluates three pass/fail criteria at every simulation step. Let $\lambda_G^{\min}$ be the smallest eigenvalue of the Gram representative of the 4-metric, $\varepsilon_{\rm det}$ the maximum absolute metric-determinant deviation from -1, and $I_{\max}$ the largest information-density flux. The safety logic is:

- if $\varepsilon_{\rm det} > 10^{-12}$, trigger EmergencyDeterminantViolation;
- else if $\lambda_G^{\min} < 0.350000$, trigger EmergencyGramEigenvalue;
- else if $I_{\max} > I_{\rm budget}$, trigger EmergencyInformationDensity;
- else return StatusNominalPass.

The HIL checks are implemented by the following three algorithms, which are called inside the Phase A, B, and C integration loops.

Algorithm 1: Metric determinant HIL check.

1. **Input:** grid metric $g_{\mu\nu}(x^k)$, tolerance $\epsilon = 10^{-12}$.
2. For each cell (i, j, k):
   (a) compute $\det g_{\mu\nu}(x^k)$;
   (b) if $|\det g + 1| > \epsilon$, return EmergencyDeterminantViolation.
3. Return StatusNominalPass.

Algorithm 2: Gram eigenvalue HIL check.

1. **Input:** Gram representative $\tilde{g}_{\mu\nu}(x^k)$, threshold $\lambda_0 = 0.350000$.
2. For each cell (i, j, k):
   (a) compute $\lambda_{\min}\big[\tilde{g}(x^k)\big]$;
   (b) if $\lambda_{\min} < \lambda_0$, return EmergencyGramEigenvalue.
3. Return StatusNominalPass.

Algorithm 3: Information-density HIL check.

1. **Input:** maximum local information density $I_{\max}$, budget $I_{\rm budget}$.
2. If $I_{\max} > I_{\rm budget}$, return EmergencyInformationDensity.
3. Return StatusNominalPass.

The HIL status string returned at each step is therefore a deterministic function of the metric, Gram, and information-density audits, and it is logged by the CAD engine alongside the flight-phase telemetry.

Appendix A: Generated LaTeX macro reference

Running `shbt-warp-sim` (or the equivalent `python-mshbt_warp.cli` entry point) writes `sim_results.tex`, which is loaded by `main.tex` before the document begins. The exported interface is listed in Table III. Alias pairs are retained for compatibility with earlier drafts.

TABLE III. Dynamically Injected Simulation Macro Exports and Audit Benchmarks.

LaTeX Macro	**Value / Benchmark Output**
`\SimOutputCanonicalBranch`	$(26, 8, 312)$
`\SimOutputFramingDefect`	0.000000000000
`\SimOutputDarkResidual`	2.300805139659
`\SimOutputDarkCompleted`	3.300805139659
`\SimOutputEntropyDebt`	0.137533547486
`\SimOutputWarpVelocity`	$1.071186351\,c$
`\SimOutputOperationalPowerMW`	142.08 MW
`\SimOutputSaturatedBitBudget`	$3.312593327986 \times 10^{122}$ bits
`\SimOutputLocalMemoryBits`	1.202481×10^{72} bits

For example, a single simulation run dynamically supplies the coupled benchmark tuple

$$\left(\frac{v_{\mathrm{eff}}}{c}, P_{\mathrm{op}}(10\,\mathrm{m}), \Delta_{\mathrm{fr}}, \lambda_G^{\min}\right) = (1.071186351, 142.08\,\mathrm{MW}, 0.000000000000, 0.358567865584)\,. \tag{A1}$$

If `sim_results.tex` is absent, `main.tex` defines matching fallback macros so that the manuscript remains compilable; regenerated simulation output takes precedence whenever it is present. The CAD flight engine exports additional macros in `cad_sim_results.tex`; Table IV extends the reference to include these dynamic quantities.

TABLE IV. Dynamic CAD engine macro exports.

LaTeX Macro	**Value / Benchmark Output**
`\CADCanonicalBranch`	$(26, 8, 312)$
`\CADFramingDefect`	0.000000000000
`\CADUnitarityResidual`	$2.220446049250e - 16$
`\CADPhaseJitterOk`	true
`\CADThermalDecoherenceOk`	true
`\CADIntegerLockOk`	true
`\CADStinespringRatio`	0.697043612789
`\CADDeltaMod`	0.137533547486
`\CADMetricDeterminant`	-1.000000000000

[1] N. Toro, *Static Holographic Boundary Theory (SHBT) precision reference implementation*, GitHub repository, `https://github.com/sys1own/shbt-precision`.
[2] M. Alcubierre, “The warp drive: hyper-fast travel within general relativity,” *Class. Quantum Grav.* **11**, L73–L77 (1994).
[3] J. M. Maldacena, “The large N limit of superconformal field theories and supergravity,” *Adv. Theor. Math. Phys.* **2**, 231–252 (1998), arXiv:hep-th/9711200.
[4] G. ’t Hooft, “Dimensional reduction in quantum gravity,” arXiv:gr-qc/9310026 (1993).
[5] J. D. Bekenstein, “Universal upper bound on the entropy-to-energy ratio for bounded systems,” *Phys. Rev. D* **23**, 287 (1981).